Boini Abhiram

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LinkedIn | GitHub | Kaggle

Education

International Institute of Information Technology, Naya Raipur (IIIT-NR) Expected Apr 2028
B.Tech, Data Science and Artificial Intelligence | ***CGPA: 8.70/10.0*** *(Sem 5)* *Naya Raipur, India*

Class XII: 96.9% | JEE Main: 97.7 percentile | Class X: 95.6%

Research — Quantitative Finance (Working Papers, in preparation)

Cross-Asset Stress-Flow Index for Early Warning of Market Drawdowns | Sole author Manuscript

- Designed the **Stress Flow Index (SFI)**, a closed-form, estimation-free network statistic aggregating directed, correlation-weighted stress transfer across a **21-asset cross-class panel** (equities, EM, commodities, FX, rates, credit; 2000–2024).
- Established the honest, bounded claim via **walk-forward evaluation**: the SFI does *not* beat realised volatility on same-day crash discrimination, but **leads an asset's own volatility ahead of idiosyncratic drawdowns in 64% of 145 episodes** — significant under **block-bootstrap** and cluster-level inference ($p = 0.002$) and validated out-of-sample in both sample halves.
- Benchmarked head-to-head against the Diebold–Yilmaz connectedness index and Billio et al. Granger-causality network; showed near-orthogonality ($|\rho| \leq 0.16$) and incremental information beyond volatility (likelihood-ratio test, $p = 0.03$).

Jumps vs. Stochastic Volatility: Out-of-Sample Option-Pricing Comparison on NSE Nifty 50 Options | Sole author Manuscript

- Implemented four pricing models from scratch — **Black–Scholes–Merton, CRR binomial tree, Merton jump-diffusion, and Heston** (semi-closed form, Little-Heston-Trap characteristic function, Gauss–Legendre quadrature) — and benchmarked on **115,709 NSE Nifty 50 option records** across pre-COVID, COVID-shock, and recovery regimes.
- Avoided circular implied-volatility logic by pricing with trailing realised volatility and scoring against traded prices; tested all model gaps with **bootstrap confidence intervals and the Diebold–Mariano test** under a strict chronological out-of-sample calibration.
- Found a **regime-dependent answer**: Heston (stochastic vol) minimises out-of-sample error in calm regimes (40.4% / 18.2% RMSE reduction vs. BSM), while Merton (jumps) wins the COVID crash (27.7%), all at $p < 0.001$; used BSM–CRR equivalence as a numerical-correctness control.

Quantitative Finance Projects

Statistical Arbitrage / Pairs Trading (NSE Equities) | Python, statsmodels github.com/Abhi241-bot/Quantfin1

- Built a cost-aware pairs strategy on TCS/INFY selected by **Engle–Granger cointegration** ($p = 0.0002$, Bonferroni-cleared) with ADF and Johansen confirmation; estimated hedge ratio and spread parameters on **train only and froze them**, with one-bar signal lag and 25 bps round-trip costs.
- Reported results **honestly**: in-sample Sharpe 0.97 decaying to −0.72 out-of-sample, and diagnosed the cause via **rolling cointegration, a Kalman-filter time-varying hedge ratio, and 9-fold walk-forward validation** (aggregate OOS Sharpe ≈ 0) — showing in-sample significance did not imply tradability.
- Added a cost-sensitivity sweep proving the OOS failure was a signal failure, not a cost artifact (loss persists at 0 bps), plus a two-pair portfolio quantifying the limits of naive equal-weight diversification; 27-test `pytest` suite enforcing look-ahead invariants.

Options Pricing & Implied Volatility Surface | Python, SciPy github.com/Abhi241-bot/Quantfin2

- Implemented **Black–Scholes (with full Greeks), a CRR binomial tree, and Monte Carlo** pricers that agree to within MC error (three-way agreement as the correctness proof); validated via put-call parity and finite-difference Greeks.
- Backed implied volatilities out of a real SPY option chain with a Brent root-finder and documented the **downward equity skew** (+9.6% at 7d → +5.1% at 88d) and term structure — framed as the market correcting the flat-vol assumption.
- Built a **delta-hedging simulation** showing hedging-error standard deviation scaling as $1/\sqrt{n}$ (Boyle–Emanuel) and turning negative under transaction costs — demonstrating the BS price as the cost of replication.

Multi-Factor Long-Short Equity Strategy (Nifty 100) | Python, pandas github.com/Abhi241-bot/Quantfin3

- Constructed **momentum, low-volatility, and value** factors as point-in-time cross-sectional z-scores; ranked into a dollar-neutral long-short book, rebalanced monthly with turnover-based costs and one-period signal lag (verified by hand-calc).

- Reported a **negative net result honestly** and used a point-in-time **beta hedge** to show the apparent +3.1% static alpha collapsed to +0.7% once market exposure was removed — a rigorous, fully attributed teardown rather than a flattering headline.
- Disclosed survivorship bias and fundamentals-lag limitations; full per-factor attribution, alpha/beta/IR vs. benchmark, and Information-Coefficient decay analysis.

Experience

Research Intern — IIT (BHU) Varanasi May 2026 – Present
Urban morphology via multi-modal semantic segmentation of satellite imagery *Varanasi, India*

- Developing deep-learning segmentation models to classify urban land-use, building footprints, road networks, and green cover across Tier-II Indian cities.
- Applying multi-modal fusion of optical and SAR satellite data to improve segmentation accuracy in heterogeneous urban environments.

AI Data Analyst (Remote Intern) — Rochester Institute of Technology Dec 2025 – Jan 2026
Student-engagement modelling and retention analytics *New York, USA (remote)*

- Built a churn classifier on **imbalanced** engagement data, prioritising F1 and precision-recall over accuracy given the low event base rate, with cross-validation to avoid overfitting.
- Applied time-series and statistical analysis to model behaviour across engagement touchpoints; delivered Tableau dashboards for stakeholder decisions.

AI/ML Engineer — Dept. of Forests, Chhattisgarh Government Dec 2025 – Present
Production voice-based logging system for a government zoo *Raipur, India*

- Shipped a production Hindi voice-logging pipeline (Deepgram ASR + Gemini LLM summarisation) with role-based inventory, medication, and veterinary modules — end-to-end from audio ingestion to database.

AI/ML Engineer Intern (Remote) — Infosys Springboard Oct 2025 – Dec 2025
Real-time facial-attribute analysis *Remote*

- Designed custom CNNs for smile/age estimation and an optimised inference pipeline (Haar Cascade + CLAHE), cutting processing latency by 40%.

Technical Skills

Programming	Python (primary), C++, SQL, R, C, Java, MATLAB
Quant & Stats	Time-series econometrics (cointegration, ADF, stationarity), options pricing (Black–Scholes, binomial trees, Monte Carlo, Merton jump-diffusion, Heston), Greeks, implied volatility, factor models, portfolio construction, backtesting, hypothesis testing, bootstrap inference, Diebold–Mariano test, Kalman filtering
Libraries	NumPy, pandas, SciPy, statsmodels, scikit-learn, PyTorch, TensorFlow/Keras, Matplotlib
ML / DL	Supervised/unsupervised learning, CNNs, RNNs, cross-validation, hyperparameter optimisation, feature engineering
Tools	Git/GitHub, Linux, Jupyter, Docker, `pytest`, REST APIs, Streamlit

Certifications & Achievements

Certifications: Completed a Quantitative Finance course (finance fundamentals & derivatives) | Kaggle Expert (Notebooks & Datasets) | IBM Data Science Professional | Oracle Cloud Infrastructure 2025 Data Science Professional | Google Gen AI Academy 2.0 | Intel AI Fundamentals.